Previous Myocardial Infarction Detection by Q Wave Amplitude

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1. Introduction

Myocardial infarction (MI), commonly known as a heart attack, occurs when blood flow to the heart is obstructed, causing damage to the cardiac tissue. Detecting previous myocardial infarctions-once or multiple times- is critical for evaluating long-term cardiovascular risk and preventing future cardiac events. A history of MI, particularly when unrecognized, significantly increases the likelihood of complications such as recurrent infarctions and other cardiovascular events, or death [1]. The Q wave, an essential component of the electrocardiogram (ECG), is a well-established marker of myocardial necrosis. Abnormal Q wave deflections from the isoelectric line are indicative of damage caused previously and are central to our goal of diagnosing prior myocardial infarctions.

One of the major challenges in detecting previous MIs is the occurrence of silent myocardial infarctions (SMIs). These events lack the properties of a heart attack, such as chest pain or shortness of breath, making them particularly difficult to diagnose [2]. SMIs are prevalent in high-risk populations, including the elderly and individuals with diabetes, and often go unnoticed until complications arise [2]. Identifying previous MI through non-invasive methods like ECG is especially critical in such cases to facilitate timely intervention and improve long-term outcomes.

Initially in our project, the detection of previous MI relied heavily on threshold-based systems. The system evaluated Q wave amplitudes and classified them based on predefined thresholds. While simple and computationally efficient, threshold-based methods are inherently limited by their inability to account for inter-individual variations in ECG presentations, noise in the data, and overlapping features between normal and pathological states. And as we used this method for our project, due to the reasons listed, we could not get any valuable classification result. The failure of threshold-based approaches led us to machine learning techniques, which are more robust and efficient.

In this study, we aimed to improve the detection of previous myocardial infarction by applying machine learning techniques, specifically a Support Vector Machine (SVM) model with utilizing leave one out cross-validation (LOOCV) approach. SVM models are well-suited for this task due to their ability to handle high-dimensional data and identify subtle patterns that threshold method cannot discern. Our approach involves extracting features from the Q wave, using these features as inputs to the SVM classifier, and optimizing the model to enhance diagnostic performance. Incomparable to the threshold method, we report that our SVM model achieved a significant improvement, attaining an average accuracy of 85%.

2. Literature Review

As we did not collect our own data, the internet was searched for ECG databases, which hold information about myocardial infarction. 5 candidate databases were selected for their size, readability and medical records. As we needed a previous myocardial infarction story from the patients, we found that only PTB databases (PTB, PTB-XL, PTB-XL+) match the criteria, and we used PTB as our final dataset. PTB Diagnostic ECG Database was acquired from 290 subjects in 2000 using high-resolution 15-lead ECGs, totaling 549 records. 52 subjects were from the healthy control group and 148 subjects had myocardial infarction, including previous ones. The dataset was first categorized into groups (Controls, Previous MI, Two Previous MI, Acute MI) by writing a sorting code on Python. Acute MI was not to be used in our classification as acute myocardial infarction could distort the Q-wave amplitudes further than we would like it to. The folders and their patient counts are as follows:

Controls:52, One_Previous_MI: 21, Two_Previous_MI: 3

As it can be seen, the dataset is not formed of many patients. And, due to data scarcity, we ruled out detecting more than one MIs in a patient, only whether they had a previous one or not. After our initial results of q-wave amplitude extraction, we decided not to use 7 patients from each group due to them being outliers and distorting the classification a lot.

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3. Dataset and Methodology

The patient data obtained from the patients were processed on MATLAB. First, 12 lead measurements with 1 kHz were taken as a basis. After the ECG data was read, it was filtered with a 0.5-40 Hz bandpass filter and then normalized.

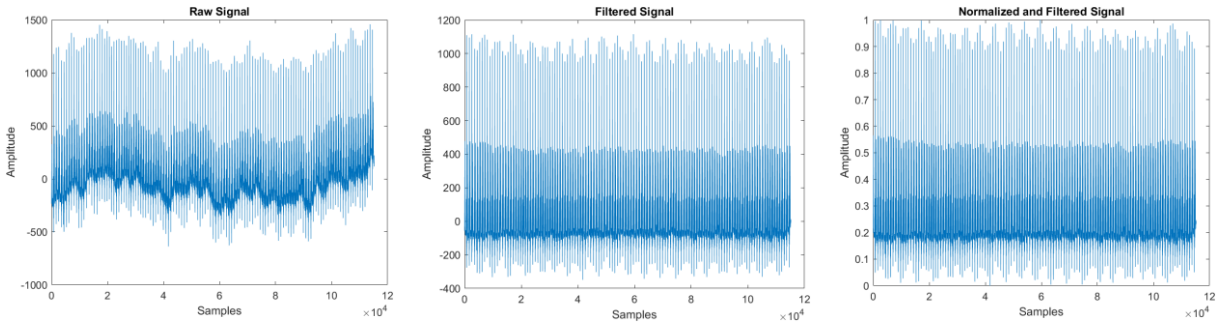


Fig.1 – Raw, filtered and normalized ECG signals plotted in Matlab.

After pre-processing, Q peak detection was started. The most prominent part of the ECG data, R peaks, was used for Q peak detection. Based on the peaks remaining above the determined threshold, all R peaks in the data were found with their locations (Sample index). Q peaks are caused by falling below the isoelectric line due to septal depolarization before the R peaks. For this reason, a 50 ms window was opened before the R peak for their detection and the maximum negative peaks within this window were taken as basis.

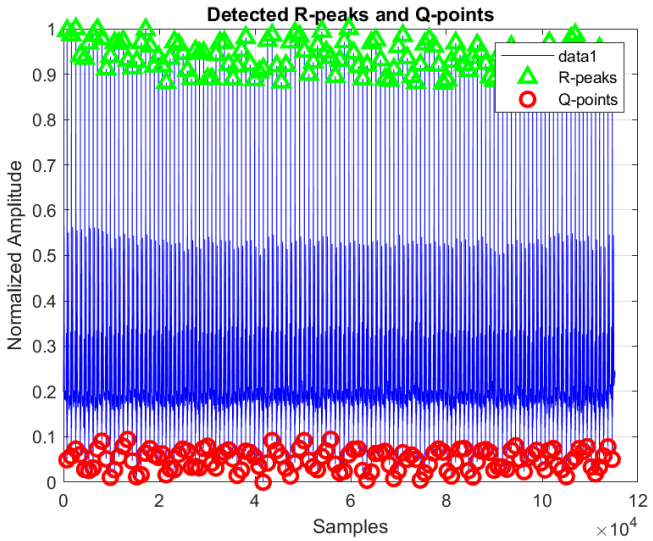


Fig.2 – Feature extraction step of R and Q peaks.

The Q peaks detected with this method were made for all instances in each lead. In order to calculate the average from the Q peaks detected for a patient, 2 different methods were followed: The average of all leads and the use of only lead 2. Q waves appear stronger in some leads than in others. This allows the detection of different scenarios based on the status of Q waves in different leads. In this study, Lead 2 was determined as the Q amplitude detection tool due to the easier detection of abnormal Q waves and the fact that it appears more stable in the data, but it was not used for all patients. The reason for this is that ECG data are not stable for every patient. While examining the ECG data in terms of suitability, it was observed that some leads had unstable measurements in some patients. This may be due to measurement problems such as incorrect placement of the lead or patient movements etc.. In order to account for this situation, the standard deviation of each lead was calculated within itself. Then, the common standard deviation of all leads was compared with the standard deviation of lead 2, and it was determined whether Lead 2 or the average of all leads would be used for the Q amplitude average value of the examined patient.

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Processing C:\Users\aykut\OneDrive\Masaüstü\ELEC447\For_project\Controls_Corrected\patient240\s0468_re.dat
Chose Lead II for C:\Users\aykut\OneDrive\Masaüstü\ELEC447\For_project\Controls_Corrected\patient240\s0468_re.dat with mean Q amplitude: 0.0741
Processing C:\Users\aykut\OneDrive\Masaüstü\ELEC447\For_project\Controls_Corrected\patient241\s0469_re.dat
Chose Lead II for C:\Users\aykut\OneDrive\Masaüstü\ELEC447\For_project\Controls_Corrected\patient241\s0469_re.dat with mean Q amplitude: 0.0311
Processing C:\Users\aykut\OneDrive\Masaüstü\ELEC447\For_project\Controls_Corrected\patient241\s0470_re.dat
Chose Lead II for C:\Users\aykut\OneDrive\Masaüstü\ELEC447\For_project\Controls_Corrected\patient241\s0470_re.dat with mean Q amplitude: 0.0391

Chose All-Leads Average for C:\Users\aykut\OneDrive\Masaüstü\ELEC447\For_project\Only_previous_Corrected\patient149\s0202bre.dat with mean Q amplitude: 0.6109
Processing C:\Users\aykut\OneDrive\Masaüstü\ELEC447\For_project\Only_previous_Corrected\patient183\s0175_re.dat
Chose All-Leads Average for C:\Users\aykut\OneDrive\Masaüstü\ELEC447\For_project\Only_previous_Corrected\patient183\s0175_re.dat with mean Q amplitude: 0.3878
Processing C:\Users\aykut\OneDrive\Masaüstü\ELEC447\For_project\Only_previous_Corrected\patient189\s0309lre.dat
Chose All-Leads Average for C:\Users\aykut\OneDrive\Masaüstü\ELEC447\For_project\Only_previous_Corrected\patient189\s0309lre.dat with mean Q amplitude: 0.4840
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Fig.3 – The average of mean Q-wave amplitudes of all-leads, and only for Lead II.

With this method, all Q peak amplitude values were determined relatively and the data was stored labeled as Control or Previous_MI. Then, a threshold was determined from taking the mean Q-wave amplitudes of Control and Previous_MI groups, and averaging those values. As simple as it is, the threshold was tested for a couple of test groups, but no significant results were observed, as the accuracy of the method did not fare better than the odds for a coin toss.

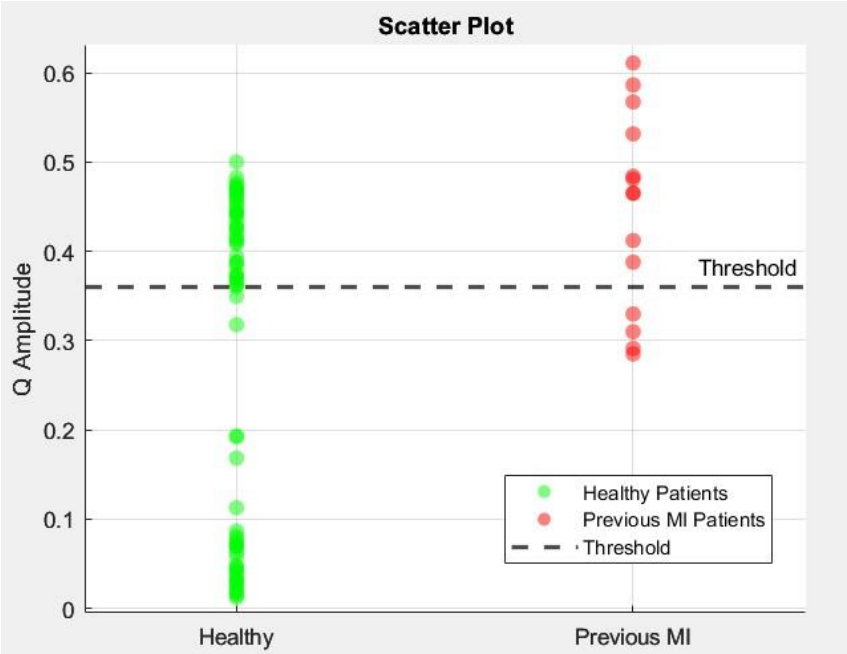


Fig.4 – Mean Q-wave amplitudes of patients, and controls, and threshold value plotted on Matlab.

After threshold method's failure, machine learning methods was searched and SVM was selected as the candidate. Afterwards, the data was shifted in a loop and divided into training and test sets to do Leave-One-Out Cross-Validation. The train data was used to train the SVM using the Radial Basis Function Kernel and in this way, all the data was used to train. After this stage, the LOOCV accuracy was calculated with the predictions during SVM training, and the Prediction results were printed to the terminal.

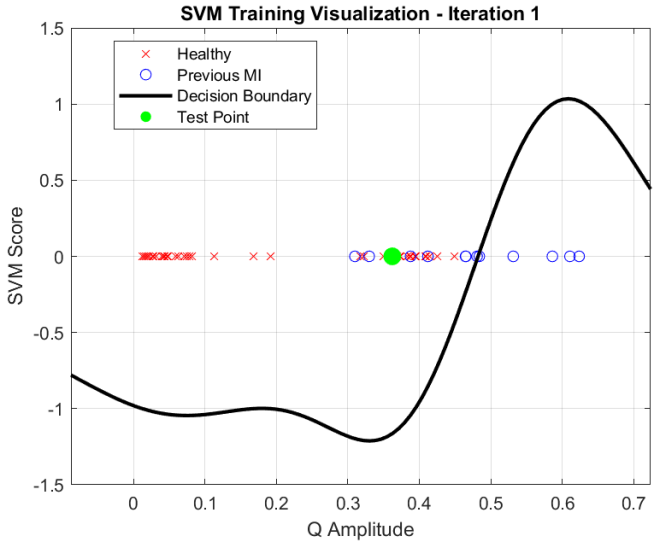


Fig. 5 – Visualization of the training of SVM method.

4. Results and Discussion

Support vector machine model was used based on Q wave amplitudes extracted from ECG data and model performance was evaluated using Leave One Out Cross Validation. As a result of this evaluation, 89.83% classification accuracy was obtained for all patients.

For the control group (healthy patients), the model consistently predicted their genuine class with a high degree of accuracy, showing strong agreement with the true labels in the majority of cases. All samples from the control group were appropriately categorized. However, several misclassifications were noted for the group that had previously experienced a myocardial infarction (MI). In particular, files such as s0201_re.dat, s0335lre.dat, and s0175_re.dat were categorized as controls even though their true labels indicated MI. This may be due to situations such as the possibility of patients having different conditions that will affect their heart conditions or undiagnosed heart conditions.

The decision boundary behavior of the first iteration is shown to visualize the SVM training in model learning. For each Leave One Out Cross Validation step, the SVM boundary was adapted to the current training data, and the data was tested with the test subject. The resulting decision boundary provided a model with an average accuracy of 89%. Classification results were generally successful, but for better classification, different features or larger datasets, especially for One Previous Myocardial Infarction, could be used.

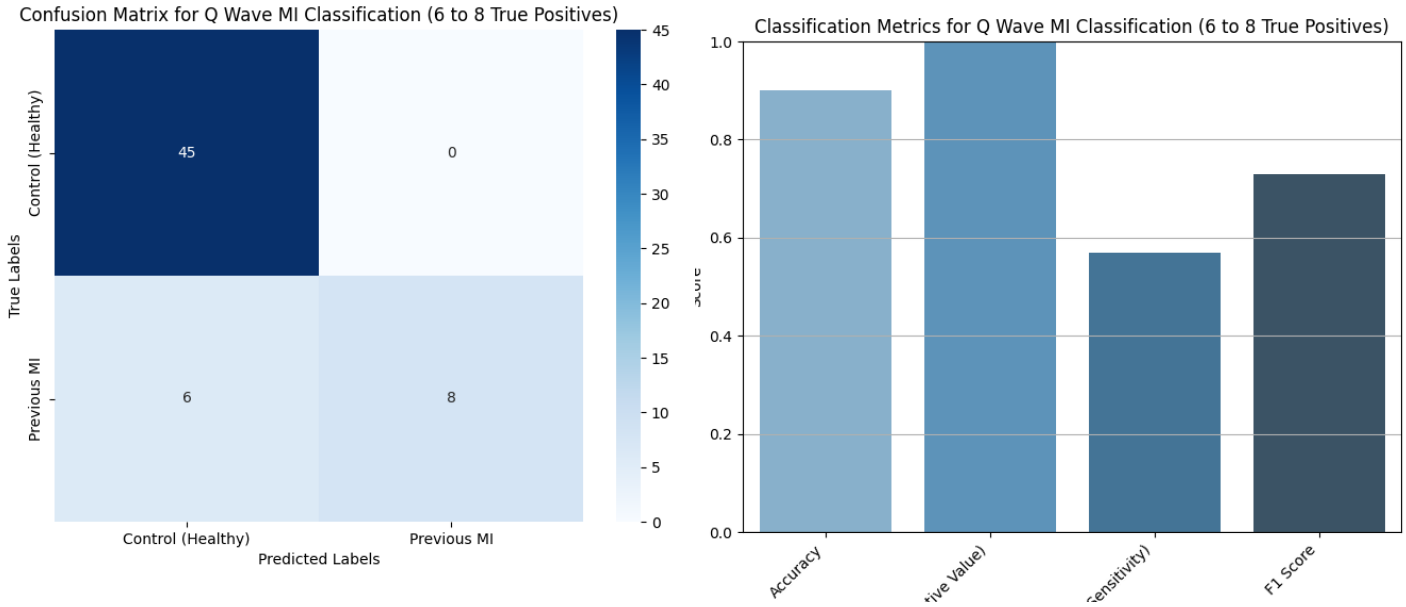


Fig. 6 – Confusion matrix and classification metrics of Support Vector Machine algorithm.

7. Conclusion

This study focused on the detection of previous myocardial infarction by analyzing Q wave deflections in ECG data. While our initial threshold-based approach was computationally efficient, it proved inadequate due to variability in ECG signals and insufficient classification accuracy. Consequently, we transitioned to a Support Vector Machine model, achieving an accuracy of ~90% using Leave-One-Out Cross-Validation. But, our study also showed the limitations of using a relatively small dataset, particularly for detecting multiple prior MIs. Expanding the dataset and incorporating additional features could further enhance the classification accuracy, but more importantly Recall and F1 score.

All in all, our work demonstrates the potential of machine learning in advancing cardiovascular diagnostics, especially in identifying silent myocardial infarctions, which often go undetected using traditional methods. By improving detection accuracy, this approach contributes to better risk assessment, early intervention, and ultimately improved patient outcomes.

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8. References

[1]Li, S., Peng, Y., Wang, X., Qian, Y., Xiang, P., Wade, S. W., ... & Handelsman, Y. (2019). "Cardiovascular events and death after myocardial infarction or ischemic stroke in an older Medicare population." *Clinical cardiology*, *42*(3), 391-399.

[2]Valensi, P., Lorgis, L., & Cottin, Y. (2011). "Prevalence, incidence, predictive factors and prognosis of silent myocardial infarction: a review of the literature." *Archives of cardiovascular diseases*, *104*(3), 178-188.